Benchmarking Open Instruction-Tuned LLMs for Greek NLP: A Reproducible Multi-Task Evaluation

Anonymous Author(s)

ABSTRACT

We present a reproducible benchmark for evaluating open instruction-tuned large language models on Greek NLP. The benchmark covers grammatical error correction, intent classification, legal document classification, machine translation, named entity recognition, part-of-speech tagging, and summarization. We evaluate five open models under a deterministic capped-test protocol that preserves ordering and exact reproducibility while reducing the runtime burden of very large test sets. Across nine benchmark segments, `gemma2:9b` ranks first overall on normalized quality, while `aya-expanse:8b` is strongest on legal classification and machine translation, and `llama3.1:8b-instruct` performs best on summarization. The paper is paired with repository-native tables, vector figures, appendix-level task analyses, and combined prediction exports so that benchmark execution and manuscript preparation remain tightly coupled.

CCS CONCEPTS

• **Computing methodologies** → **Natural language processing**; *Language resources*; • **Information systems** → Data analytics.

KEYWORDS

Greek NLP, benchmarking, large language models, reproducibility, evaluation

1 INTRODUCTION

Broad evaluation has become central to modern language model research because single-task scores rarely capture deployment-relevant trade-offs in robustness, latency, and transfer [1, 3, 4]. Yet, for Greek NLP, evaluation resources still remain fragmented across task-specific datasets, metrics, and execution pipelines. This makes it difficult to compare open instruction-tuned LLMs under a shared protocol, especially when the goal is not a single downstream task but a broader view across heterogeneous settings.

Instruction tuning has made general-purpose language models more usable across many tasks without task-specific retraining [2, 5]. However, evaluating such models on Greek remains cumbersome in practice. This paper packages our evaluation workflow into a reproducible benchmark suitable for SETN 2026. The benchmark aggregates seven task families into nine reporting segments: grammatical error correction (GEC), intent classification, legal classification, machine translation into three target languages, named entity recognition (NER), part-of-speech (POS) tagging, and summarization. The runner produces per-task summaries, per-instance predictions, combined CSV outputs, HTML visualizations, and publication-facing figures.

Our contribution is practical rather than algorithmic. First, we provide a unified multi-task evaluation setup for Greek NLP with open local models. Second, we adopt a deterministic capped-test protocol that remains reproducible without relying on random sampling. Third, we expose benchmark outputs in a form that flows directly into a conference paper, including automatically generated appendix materials for task-level analysis.

2 BENCHMARK SETUP

2.1 Tasks and Models

The benchmark covers seven task families:

- GEC
- intent classification
- legal classification
- machine translation
- NER
- POS tagging
- summarization

We evaluate five open instruction-tuned models served locally through Ollama: `qwen2.5:7b-instruct`, `llama3.1:8b-instruct`, `aya-expanse:8b`, `gemma2:9b`, and `falcon3:7b-instruct`.

2.2 Deterministic Capped-Test Protocol

The full test sets are very uneven in size, with legal classification and machine translation being especially large. To keep evaluation tractable while preserving reproducibility, we use a deterministic capped-test profile that always takes the first instances in dataset order instead of random samples. The resulting capped evaluation contains:

- GEC: 175 examples
- intent classification: 436 examples
- legal classification: 500 examples
- machine translation: 500 English pairs, 500 Persian pairs, and 500 Japanese pairs
- NER: 500 examples
- POS: 456 examples
- summarization: 300 examples

This design keeps the smaller tasks intact while reducing the runtime burden of the largest tasks. It also supports exact reruns because the selected subset is deterministic. For machine translation, the benchmark now uses a shared data-wrapper path rather than a separate task-local split implementation, which eliminates an earlier source of inconsistency in language-specific counts.

2.3 Metrics

We report task-appropriate primary metrics. For classification-style tasks we use accuracy and macro-F1, with macro-F1 treated as the primary comparison metric. For GEC we report exact match and edit-distance-derived measures against the reference and source text. For machine translation we use chrF as the primary reporting metric and retain WER/CER as supplementary string-distance diagnostics. For summarization we use BERTScore-F1 as the primary semantic comparison metric and report WER/CER and ROUGE as supplementary overlap-based diagnostics. We also compute GLEU

for GEC and entity-level F1 for NER. Latency is reported as average per-instance generation time in seconds.

To compare models across heterogeneous metrics, we additionally compute a direction-aware normalized quality score per benchmark segment. Within each segment, the best model receives the highest normalized value and the worst model the lowest. These segment-normalized scores are then averaged to obtain an overall ranking. The main paper focuses on the primary metrics used for ranking, while Appendix C reports the richer per-task metric panels.

3 IMPLEMENTATION NOTES

The benchmark pipeline was adjusted in several ways to make end-to-end runs stable and paper-ready. We added combined prediction exports so that the full suite produces unified CSV files across tasks. We also changed legal classification from opaque subject identifiers to coarse-grained human-readable volume labels, which avoids the degenerate all-zero behavior observed with raw numeric class IDs. For POS tagging, we added a fallback parser that tolerates _ head values in CoNLL-U files. At the backend level, we introduced retry and backoff logic around transient Ollama failures. Finally, we generated publication-oriented reports, tables, vector graphics, and appendix-ready per-task analyses directly from benchmark outputs.

4 RESULTS

Table 1 summarizes the best model in each benchmark segment. Figure 1 visualizes normalized quality across segments, while Figure 2 shows the trade-off between quality and inference speed. Appendix C then drills into each task with one analytical table and one corresponding figure.

Four patterns stand out. First, `gemma2:9b` remains the strongest overall model in this run, ranking first on the normalized aggregate with an average quality score of 71.6 and winning GEC, intent classification, NER, POS, and two of the three machine translation segments under chrF. Second, `llama3.1:8b-instruct` rises to second overall once summarization is ranked semantically with BERTScore-F1 rather than by WER, which highlights the sensitivity of aggregate rankings to metric choice. Third, `aya-expanse:8b` remains especially competitive on cross-lingual tasks, winning legal classification, the English translation segment by WER's closest competitor metric chrF only narrowly losing to `gemma2:9b`, and taking the top summarization BERTScore-F1. Fourth, the weakest aggregate model is not uniformly weak: for example, `falcon3:7b-instruct` is often among the fastest models even when it lags in quality, which reinforces the value of reporting latency alongside task scores.

Legal classification remains the most difficult benchmark component. Even after replacing opaque identifiers with coarse-grained volume labels, the best macro-F1 is only 0.011. This suggests that the task is still demanding under zero-shot prompting and may benefit from better label descriptions, exemplars, or task-specific fine-tuning. By contrast, intent classification is comparatively mature in this setup, with `gemma2:9b` reaching 0.737 macro-F1 on 436 capped test examples. The appendix confirms that the legal task is difficult under both accuracy and macro-F1, not only under the headline metric.

Machine translation is mixed but now much more interpretable than in our earlier smaller-language slices. The capped profile evaluates 500 examples per target language, which is enough to separate models consistently while keeping runtime manageable. Using chrF as the primary metric changes the winners relative to WER: `gemma2:9b` leads English and Japanese translation, while `qwen2.5:7b-instruct` jumps to first on Persian because its chrF substantially exceeds the rest of the field despite weaker WER. This is precisely the kind of metric dependence that argues for reporting both primary and supplementary MT metrics in the appendix rather than relying on a single surface-form score.

The appendix also surfaces secondary structure in tasks where the main ranking alone would be too coarse. For NER, `llama3.1:8b-instruct` has the highest accuracy while `gemma2:9b` leads on macro-F1, suggesting differences in minority-entity handling. For summarization, the shift from WER to BERTScore-F1 changes the segment winner from `llama3.1:8b-instruct` to `aya-expanse:8b`, but only by a very small margin (0.519 vs. 0.517), which suggests that the top two models are effectively near-tied semantically even though the overlap-based and distance-based metrics tell somewhat different stories.

5 DISCUSSION

The benchmark is useful because it combines multiple Greek NLP settings in a single reproducible workflow, but the current evaluation still has limitations. The capped protocol is intentionally pragmatic rather than exhaustive. It is well suited for iterative comparison and paper-scale reporting, but it is not a substitute for full test-set evaluation when final archival numbers are required. Some metric choices are also provisional. In particular, summarization and machine translation would benefit from stronger semantic metrics and human assessment, while legal classification would benefit from richer label semantics. Even so, the appendix-level supplemental metrics already improve interpretability relative to a WER-only or macro-F1-only summary.

Despite these caveats, the benchmark already exposes meaningful structure in model behavior. Some models offer strong general performance across multiple tasks, while others are better understood as specialists for translation or summarization. This kind of cross-task perspective is hard to obtain from isolated task runs and is exactly where a unified benchmark becomes valuable.

6 REPRODUCIBILITY AND ARTIFACT STRUCTURE

The repository includes the benchmark runner, combined CSV outputs, HTML task visualizations, report generation scripts, and publication-ready assets. The capped run used in this draft is stored under the repository's completed-run hierarchy, and the article-facing tables and vector figures are generated into a dedicated `article_assets` subdirectory. This structure allows the manuscript to be regenerated directly from benchmark outputs without manual copy-paste into the paper.

7 CONCLUSION

We presented a reproducible Greek NLP benchmark draft for open instruction-tuned LLMs and used it to compare five models across

Table 1: Best-performing model per benchmark segment on the deterministic capped evaluation run (GEC=175, Intent=436, Legal=500, MT(ENG=500, FAS=500, JPN=500), NER=500, POS=456, Summarization=300), using representative metrics per task.

Task Segment	Winner	Primary Metric	Value
GEC	Llama 3.1 8B Instruct	GLEU	0.796
Intent	Gemma2 9B	Macro-F1	0.737
Legal	Aya Expanse 8B	Macro-F1	0.011
MT-ENG	Gemma2 9B	BLEU	35.073
MT-FAS	Qwen2.5 7B Instruct	BLEU	4.183
MT-JPN	Gemma2 9B	BLEU	6.227
NER	Gemma2 9B	Macro-F1	0.122
POS	Gemma2 9B	Macro-F1	0.486
Summ.	Aya Expanse 8B	BERTScore-F1	0.519

Figure 1: Normalized quality heatmap across the nine benchmark segments. Scores are normalized within each segment so that higher is always better, even when the underlying metric is lower-is-better such as WER.

seven task families and nine benchmark segments. Under a deterministic capped-test protocol, `gemma2:9b` ranks first overall, while `aya-expanse:8b` and `llama3.1:8b-instruct` emerge as strong specialists on legal classification and summarization respectively. The current pipeline is already useful for comparative evaluation, and it also provides a clean path from benchmark execution to publication assets.

Table 2: Overall ranking of the evaluated models across the nine benchmark segments in the deterministic capped benchmark run. Normalized quality is averaged over segment-level, direction-aware scores.

Model	Rank	Avg. Quality	Avg. Latency (s)	Segment Win
Gemma2 9B	1	74.5	1.48	
Llama 3.1 8B Instruct	2	59.7	1.15	
Qwen2.5 7B Instruct	3	45.0	1.37	
Aya Expanse 8B	4	43.9	1.12	
Falcon3 7B Instruct	5	30.4	1.31	

A OVERALL RANKING TABLE

The table below reports the overall model ranking induced by the normalized segment scores.

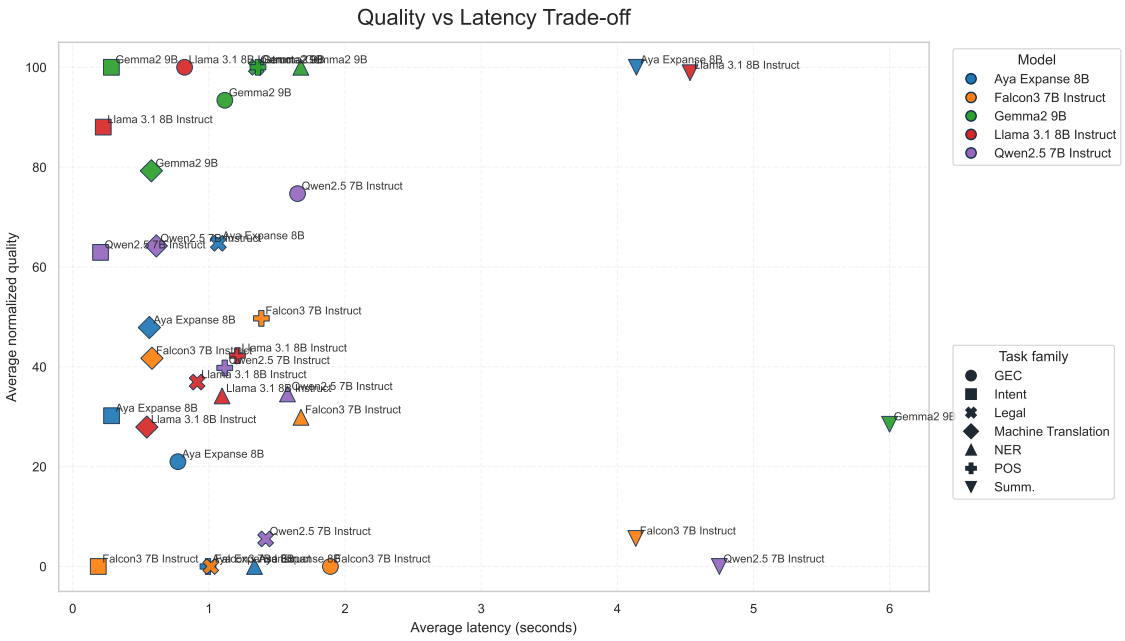

Figure 2: Quality-latency trade-off across models and task families. Each point corresponds to one model-task pair after averaging within the task family.

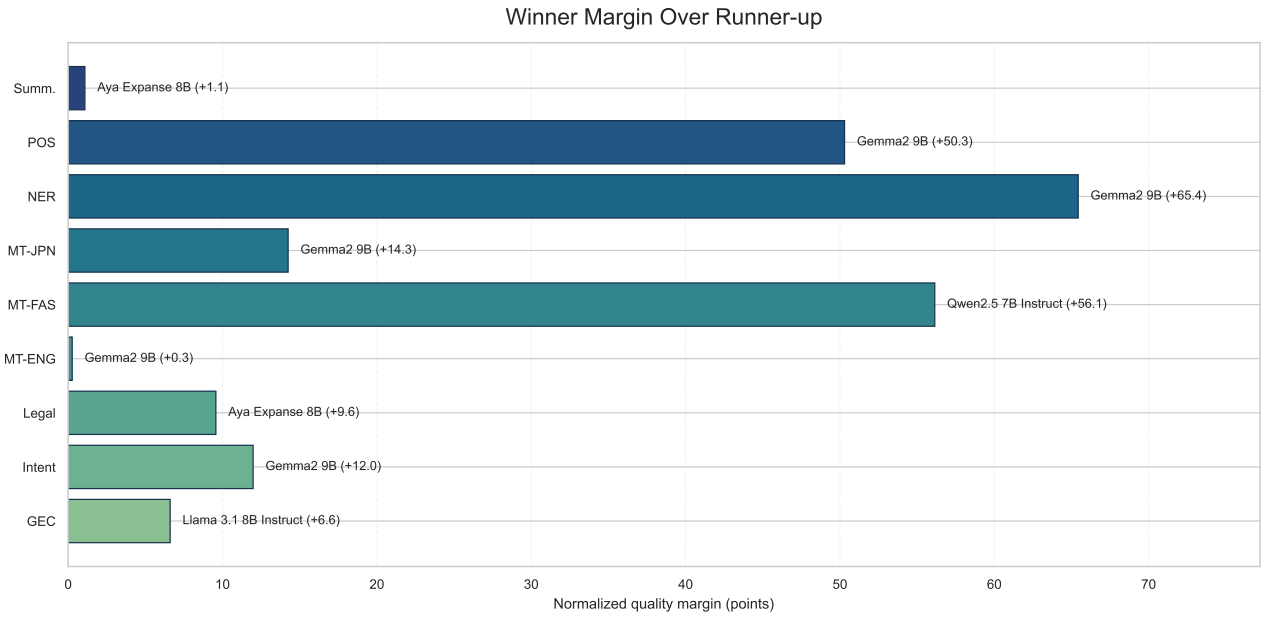

Figure 3: Normalized quality margin between the winning model and the runner-up for each benchmark segment. Large margins indicate clearer task-level separation between the top two models.

B WINNER MARGINS BY SEGMENT

C PER-TASK ANALYSIS

C.1 GEC

C.2 Intent Classification

C.3 Legal Classification

C.4 Machine Translation

C.5 NER

C.6 POS

C.7 Summarization

replacing angle-bracket placeholders with task-specific content at runtime.

Table 3: Per-model results for GEC.

Model	Exact Match	WER	Latency (s)
Llama 3.1 8B Instruct	0.120	10.489	0.822
Gemma2 9B	0.137	11.670	1.117
Qwen2.5 7B Instruct	0.069	13.324	1.651
Aya Expanse 8B	0.057	21.025	0.772
Falcon3 7B Instruct	0.006	27.049	1.892

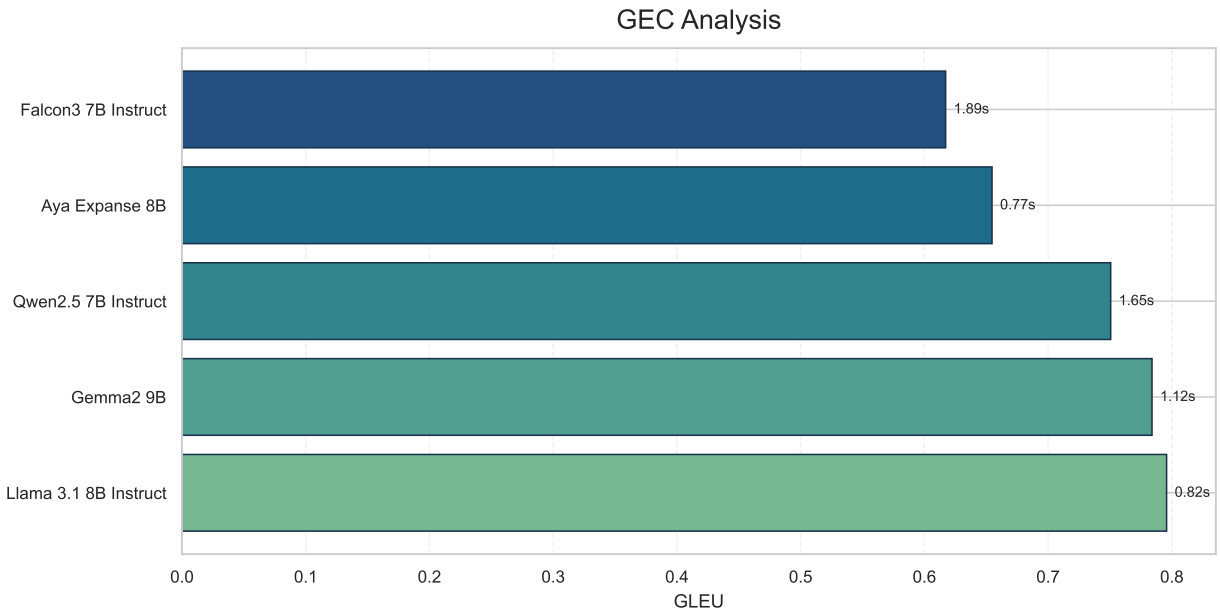

Figure 4: Per-task comparison for GEC. Bars show the primary metric and annotations show average latency.

Table 4: Per-model results for Intent.

Model	Accuracy	Macro-F1	Latency (s)
Gemma2 9B	0.862	0.737	0.284
Llama 3.1 8B Instruct	0.720	0.707	0.224
Qwen2.5 7B Instruct	0.752	0.645	0.204
Aya Expanse 8B	0.716	0.563	0.285
Falcon3 7B Instruct	0.578	0.488	0.188

Table 5: Per-model results for Legal.

Model	Accuracy	Macro-F1	Latency (s)
Gemma2 9B	0.346	0.126	1.349
Aya Expanse 8B	0.292	0.098	1.070
Llama 3.1 8B Instruct	0.182	0.076	0.914
Qwen2.5 7B Instruct	0.202	0.051	1.417
Falcon3 7B Instruct	0.122	0.047	1.014

D.1 GEC

System prompt.

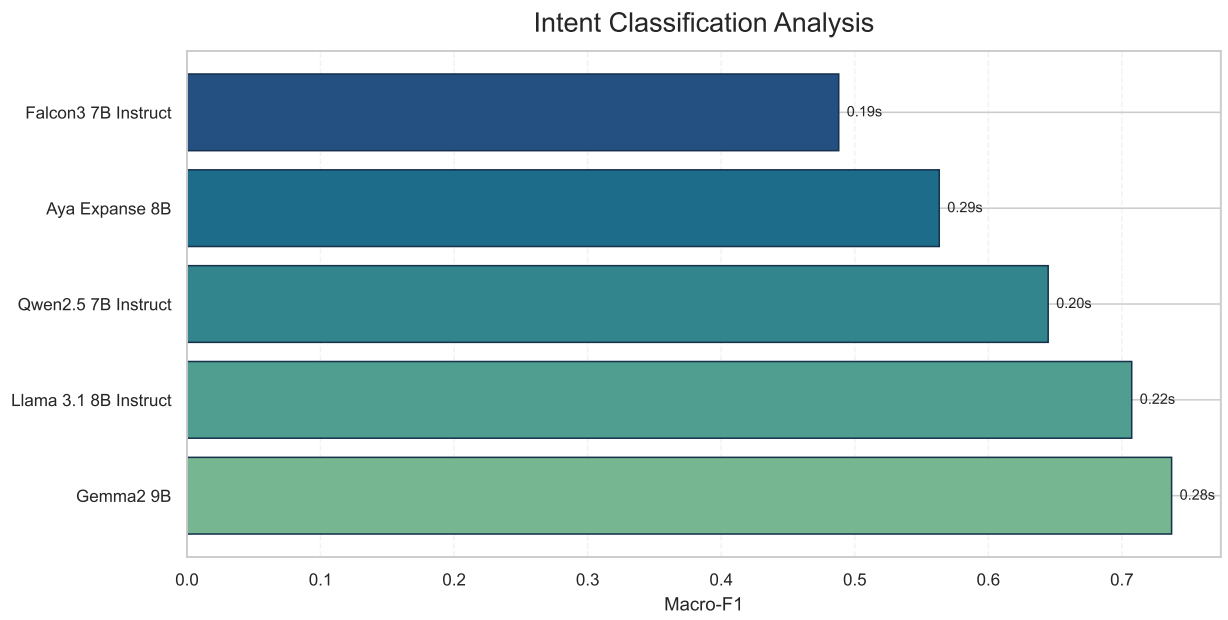

Figure 5: Per-task comparison for Intent Classification. Bars show the primary metric and annotations show average latency.

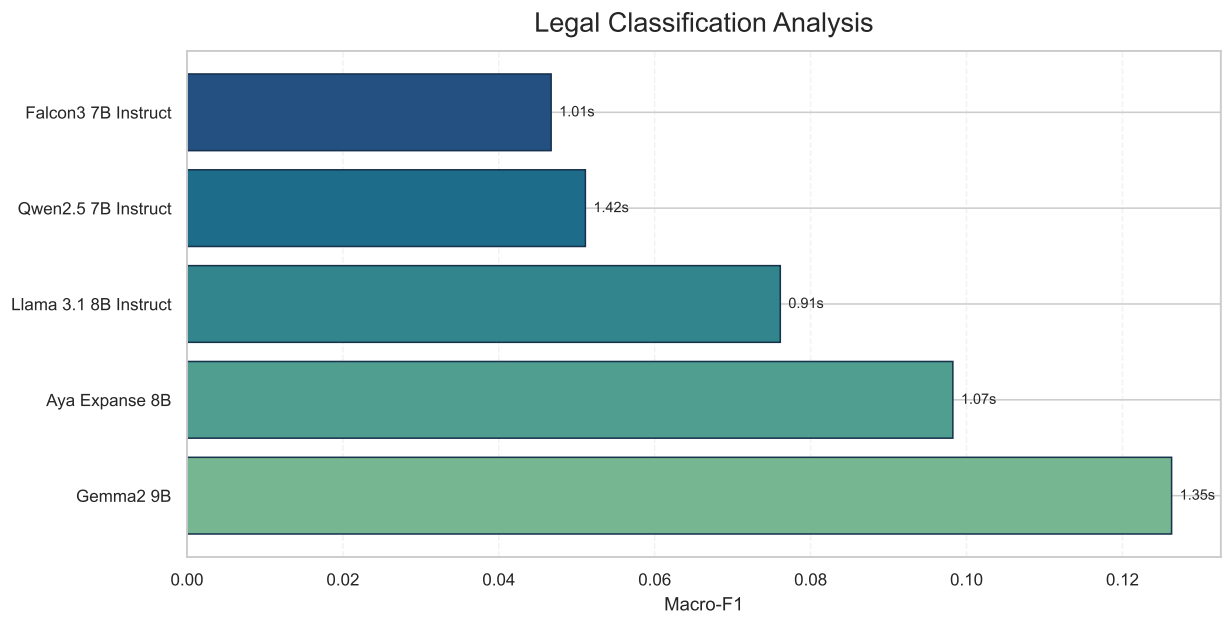

Figure 6: Per-task comparison for Legal Classification. Bars show the primary metric and annotations show average latency.

```
You are a careful grammatical error correction system for Modern Greek.
Correct the user's sentence while preserving its meaning.
Return only the corrected Greek sentence and nothing else.
```

```
Keimeno: <ORIGINAL_TEXT>
Diorthomeno keimeno:
```

Prompt template.

```
Diorthose to parakato elliniko keimeno grammatika kai orthografika.
Kratise to idio noima.
```

D.2 Intent Classification

System prompt.

Table 6: Per-model results for Machine Translation.

Lang	Model	WER	BLEU	chrF	Latency (s)
ENG	Gemma2 9B	57.915	35.073	61.781	0.569
ENG	Aya Expanse 8B	56.359	35.037	61.591	0.531
ENG	Llama 3.1 8B Instruct	64.237	32.576	60.045	0.441
ENG	Qwen2.5 7B Instruct	66.484	29.280	56.907	0.439
ENG	Falcon3 7B Instruct	77.902	21.355	49.132	0.431
JPN	Gemma2 9B	109.538	6.227	16.726	0.629
JPN	Falcon3 7B Instruct	115.930	5.665	8.895	0.900
FAS	Qwen2.5 7B Instruct	116.337	4.183	18.325	0.845
JPN	Qwen2.5 7B Instruct	168.901	3.656	12.553	0.558
JPN	Llama 3.1 8B Instruct	168.112	2.358	14.845	0.588
JPN	Aya Expanse 8B	109.067	2.279	16.533	0.610
FAS	Aya Expanse 8B	109.222	2.146	3.408	0.548
FAS	Falcon3 7B Instruct	114.451	1.984	3.124	0.416
FAS	Gemma2 9B	110.071	1.926	2.906	0.536
FAS	Llama 3.1 8B Instruct	343.614	0.555	2.058	0.604

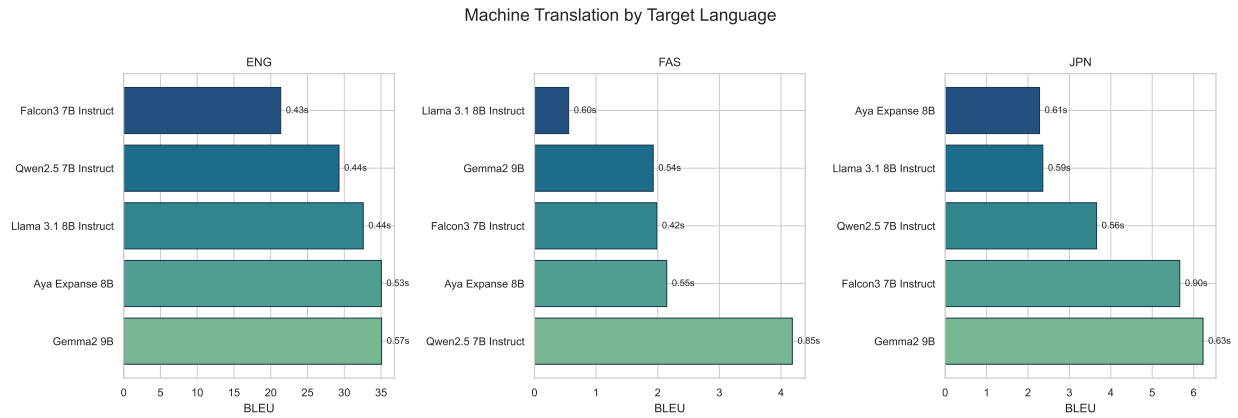

Figure 7: Per-task comparison for Machine Translation. Bars show the primary metric and annotations show average latency.

Table 7: Per-model results for NER.

Model	Accuracy	Macro-F1	Entity F1	Latency (s)
Gemma2 9B	0.796	0.122	0.043	1.675
Qwen2.5 7B Instruct	0.689	0.095	0.045	1.577
Llama 3.1 8B Instruct	0.812	0.094	0.049	1.097
Falcon3 7B Instruct	0.804	0.093	0.026	1.677
Aya Expanse 8B	0.735	0.080	0.010	1.335

```
You are a Greek intent classification system.
Return only the intent label and nothing else.
```

Prompt template.

```
Given a text, provide the intent of the text. Only the intent should be returned.
Here is the list of possible intents: <LABEL_SPACE>

Text: <TEXT>
Intent:
```

D.3 Legal Classification

System prompt.

```
You are a Greek legal text classification system.
Return only one label from the provided label set and nothing else.
```

Prompt template.

```
Given the following Greek legal text, return only its coarse-grained
Choose exactly one label from: <LABEL_SPACE>
```

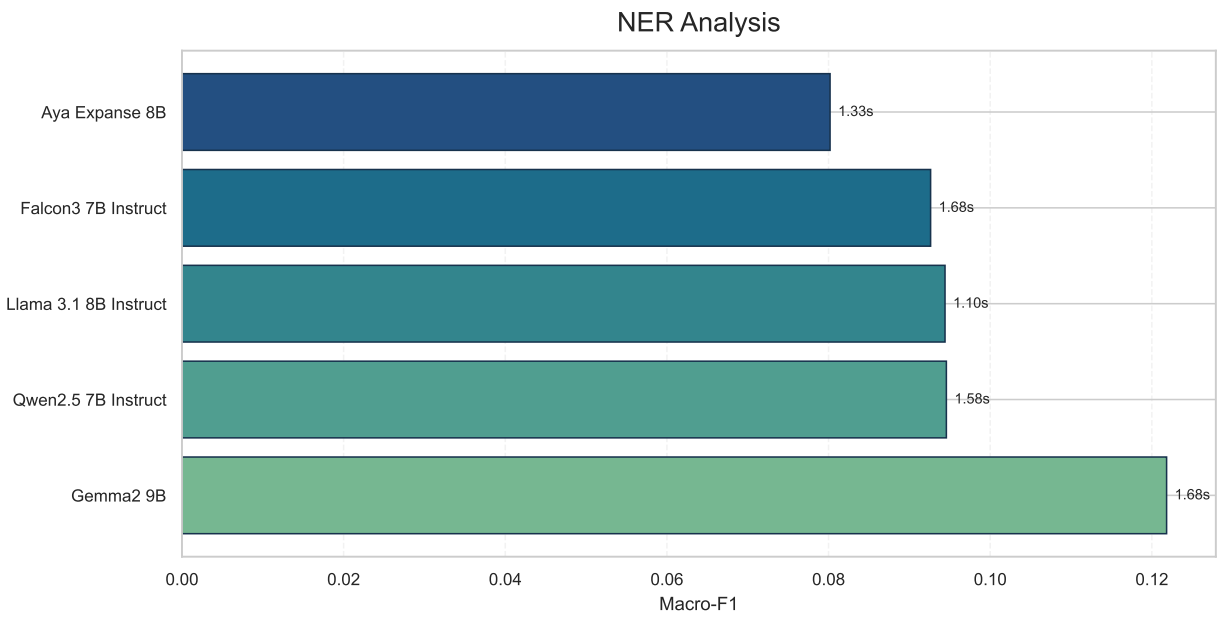

Figure 8: Per-task comparison for NER. Bars show the primary metric and annotations show average latency.

Table 8: Per-model results for POS.

Model	Accuracy	Macro-F1	Latency (s)
Gemma2 9B	0.603	0.486	1.361
Falcon3 7B Instruct	0.438	0.332	1.385
Llama 3.1 8B Instruct	0.390	0.309	1.210
Qwen2.5 7B Instruct	0.354	0.302	1.116
Aya Expanse 8B	0.229	0.180	0.993

Table 9: Per-model results for Summ..

Model	WER	ROUGE-1	ROUGE-L	BERTScore-F1	Latency (s)
Aya Expanse 8B	208.784	0.064	0.063	0.519	4.140
Llama 3.1 8B Instruct	173.585	0.081	0.076	0.517	4.534
Gemma2 9B	271.162	0.028	0.027	0.407	5.999
Falcon3 7B Instruct	235.214	0.009	0.009	0.371	4.135
Qwen2.5 7B Instruct	327.113	0.016	0.014	0.362	4.749

```
Text: <TEXT>
Label:
```

D.4 Machine Translation

System prompt.

```
You are a translation system.
Translate the user's text accurately.
Return only the translation and nothing else.
```

Prompt template.

```
Given a text in Greek, translate it to <TARGET_LANG>.
Only the translation should be returned.

Text: <SOURCE_TEXT>
Translation:
```

D.5 Named Entity Recognition

System prompt.

```
You are a Greek named entity recognition system.
Return only a Python-style list of tags and nothing else.
```

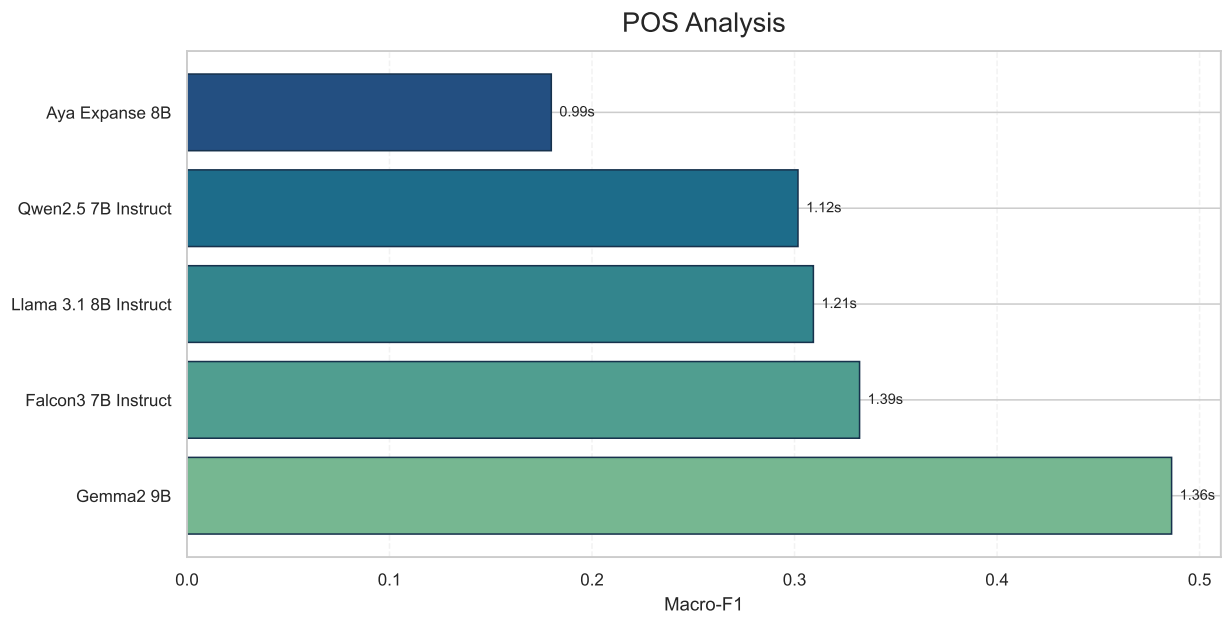

Figure 9: Per-task comparison for POS. Bars show the primary metric and annotations show average latency.

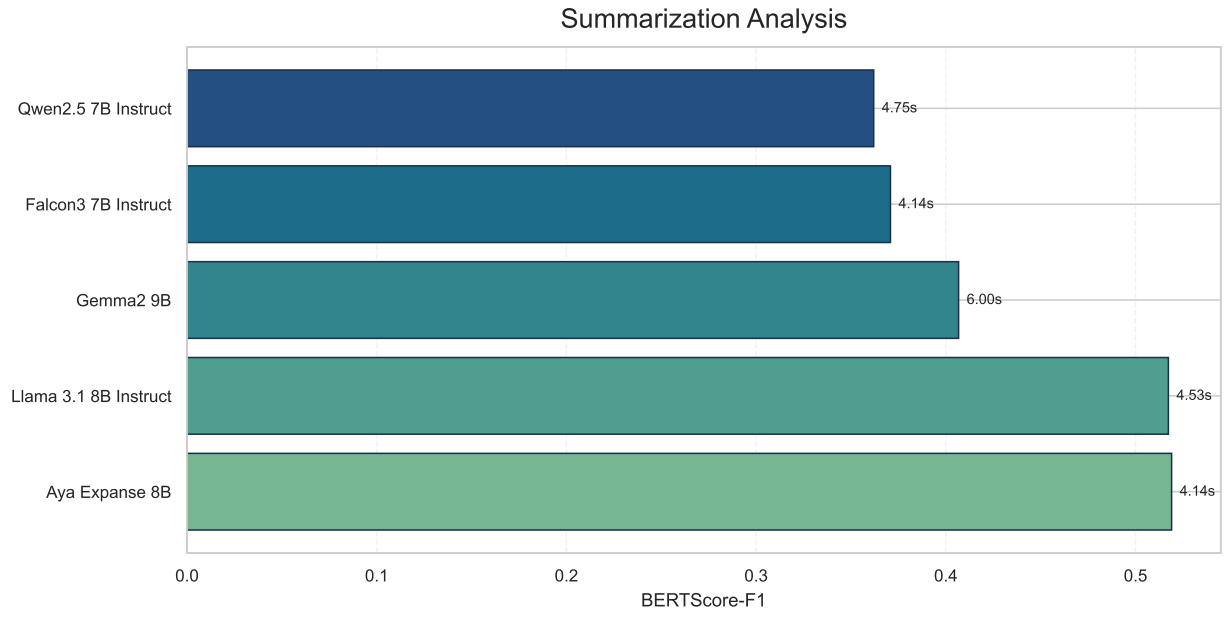

Figure 10: Per-task comparison for Summarization. Bars show the primary metric and annotations show average latency.

Prompt template.

```
You are given a tokenized Greek sentence. Return only a Python-style list of
NER tags, one tag per token.
Use only these labels: <NER_LABEL_LIST>

Tokens: <TOKEN_LIST>
Tags:
```

D.6 POS Tagging

System prompt.

```
You are a Greek POS tagging system.
Return only a Python-style list of tags and nothing else.
```

Prompt template.

```
You are given a tokenized Greek sentence. Return only a Python-style l
```

```
POS tags, one tag per token.
Use only these labels: <POS_LABEL_LIST>

Tokens: <TOKEN_LIST>
Tags:
```

D.7 Summarization

System prompt.

```
You are a Greek summarization system.
Produce a concise summary in Greek and return only the summary.
```

Prompt template.

```
Given a Greek legal text, provide its summary also in Greek.
Only the summary should be returned.

Text: <DOCUMENT_TEXT>
Summary:
```

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